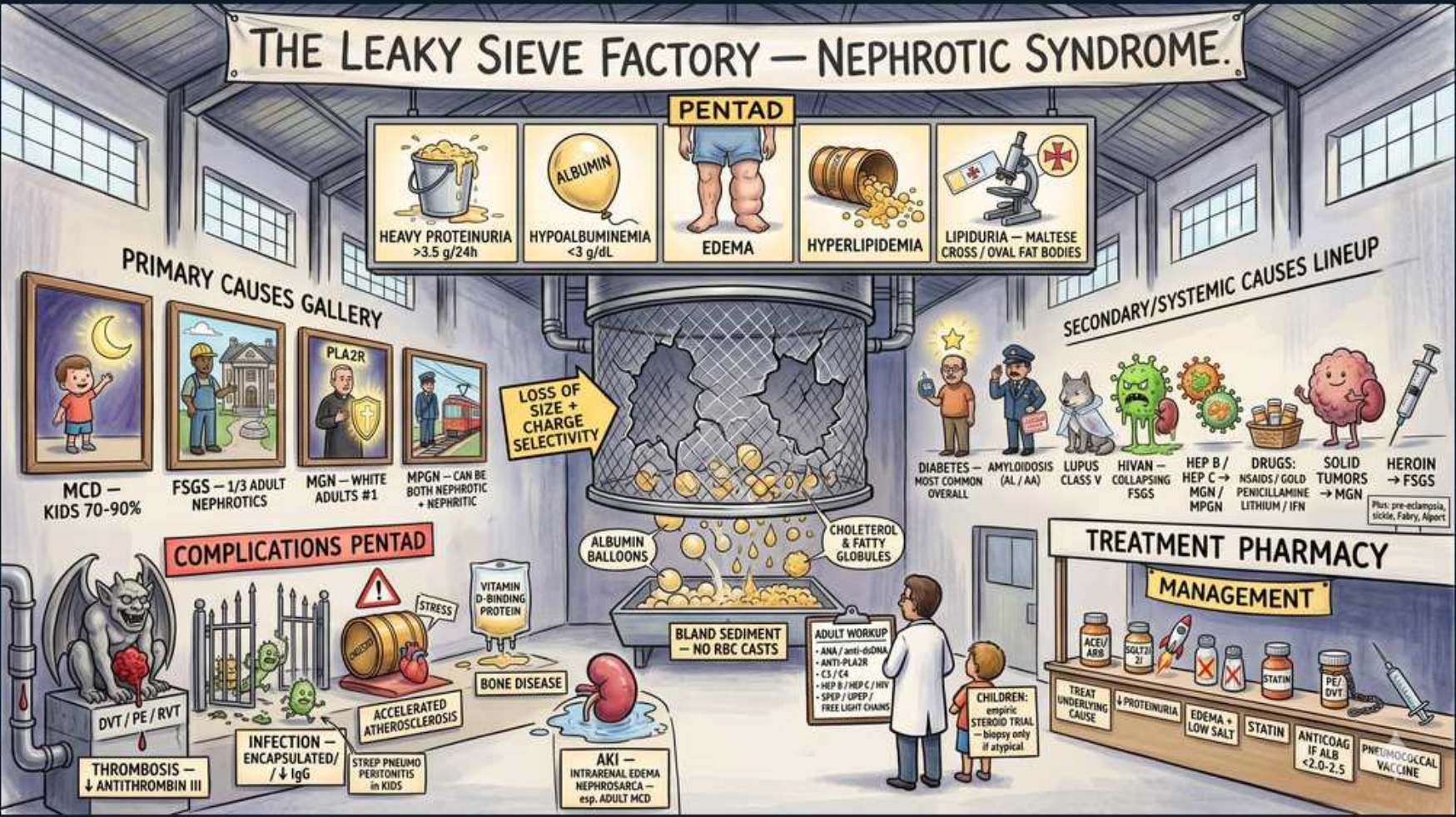


Nephrotic Syndrome Overview — The Leaky Sieve Factory



1. Heavy Proteinuria >3.5 g/24h

WHAT YOU SEE

Bucket overflowing, top pentad

WHAT IT ENCODES

Defining feature of nephrotic syndrome: protein excretion >3.5 g/day from a damaged glomerular filtration barrier losing charge/size selectivity. Driven by podocyte injury. Distinguishes nephrotic from nephritic (where proteinuria is usually <3.5 g and hematuria dominates).

BOARD TRAP

Sub-nephrotic proteinuria (<3.5 g) with hematuria and RBC casts is nephritic, not nephrotic — don't mix the syndromes.

2. Hypoalbuminemia <3 g/dL

WHAT YOU SEE

Balloon labeled hypoalbuminemia

WHAT IT ENCODES

Urinary albumin loss exceeds hepatic synthesis, dropping serum albumin <3 g/dL. Reduced oncotic pressure drives edema and stimulates compensatory hepatic lipoprotein synthesis (hyperlipidemia).

BOARD TRAP

Low albumin from nephrotic loss differs from cirrhosis (decreased synthesis) and malnutrition — check the urine protein.

3. Edema

WHAT YOU SEE

Swollen legs, pentad

WHAT IT ENCODES

Periorbital and dependent/pitting edema from low oncotic pressure plus primary renal sodium retention (ENaC activation). Can progress to anasarca. Sodium restriction and loop diuretics are first-line management.

BOARD TRAP

Nephrotic edema is often a mix of underfill (low oncotic) AND overfill (Na retention) — pure underfill is the classic but incomplete board model.

4. Hyperlipidemia

WHAT YOU SEE

Barrel of lipids

WHAT IT ENCODES

Low oncotic pressure triggers hepatic overproduction of lipoproteins and reduced clearance, raising LDL and triglycerides. Contributes to accelerated atherosclerosis. Statins are used for persistent nephrotic dyslipidemia.

BOARD TRAP

Hyperlipidemia is a consequence of hypoalbuminemia, not a primary lipid disorder — it resolves when the nephrosis remits.

5. Lipiduria — Maltese Cross / Oval Fat Bodies

WHAT YOU SEE

Microscope, fat bodies

WHAT IT ENCODES

Filtered lipoproteins appear in urine as oval fat bodies and 'Maltese cross' under polarized light, and fatty/waxy casts. Completes the nephrotic pentad. A bland sediment otherwise (no RBC casts).

BOARD TRAP

Maltese cross = lipiduria of nephrotic syndrome; RBC casts would instead point to nephritic GN.

6. Loss of Charge Selectivity

WHAT YOU SEE

Central sieve filter

WHAT IT ENCODES

The glomerular basement membrane and podocyte slit diaphragm normally repel anionic albumin. Podocyte effacement and loss of negative charge let albumin leak through — the core mechanism of selective proteinuria (esp. in minimal change disease).

BOARD TRAP

Selective (albumin-predominant) proteinuria suggests MCD/podocytopathy; non-selective (all sizes) suggests structural GBM/membranous damage.

7. MCD — Kids 70–90%

WHAT YOU SEE

Primary causes gallery, child

WHAT IT ENCODES

Minimal change disease is the most common primary cause of nephrotic syndrome in children (70–90%). LM normal, EM shows diffuse foot process effacement. Highly steroid-responsive. Associated with Hodgkin lymphoma and NSAIDs.

BOARD TRAP

In a child with nephrotic syndrome you treat empirically with steroids — biopsy is reserved for atypical features or steroid resistance.

8. FSGS — Adults #1

WHAT YOU SEE

Gallery panel, FSGS

WHAT IT ENCODES

Focal segmental glomerulosclerosis is the leading cause of primary nephrotic syndrome in adults, especially African ancestry (APOL1). Segmental sclerosis on LM, foot process effacement on EM. Variable steroid response.

BOARD TRAP

FSGS may be primary (treat with steroids) or secondary/adaptive (treat the driver + RAAS) — distinguish before immunosuppressing.

9. MGN — White Adults #1

WHAT YOU SEE

Gallery panel, membranous

WHAT IT ENCODES

Membranous nephropathy is the most common primary nephrotic cause in older white adults. Subepithelial immune deposits ('spike and dome'), anti-PLA2R antibody in ~70% of primary cases. Secondary causes: solid tumors, hepatitis B, SLE, gold/penicillamine.

BOARD TRAP

Membranous carries the highest risk of renal vein thrombosis of all glomerulonephropathies — anticoagulate if albumin very low.

10. MPGN — Both Nephrotic & Nephritic

WHAT YOU SEE

Gallery panel, MPGN

WHAT IT ENCODES

Membranoproliferative GN presents with a mixed nephrotic/nephritic picture, low complement, and 'tram-track' GBM double contours. Driven by immune complexes (hep C/cryoglobulins) or complement dysregulation (C3 glomerulopathy).

BOARD TRAP

MPGN is the classic 'overlap' lesion (nephrotic AND nephritic with low C3) — don't force it into one syndrome.

11. Secondary/Systemic Lineup

WHAT YOU SEE

Right wall, systemic causes

WHAT IT ENCODES

Major systemic causes: diabetes (Kimmelstiel-Wilson nodules — #1 overall), amyloidosis (Congo red, apple-green birefringence), lupus class V, HIV (collapsing FSGS), hepatitis B/C, drugs (NSAIDs→MCD, gold/penicillamine→MGN), solid tumors, heroin→FSGS.

BOARD TRAP

Diabetic nephropathy is the single most common cause of nephrotic syndrome overall (and of ESRD) — don't overlook it for the flashy primaries.

12. Hypercoagulability / Thrombosis

WHAT YOU SEE

Gargoyle, DVT/PE/RVT

WHAT IT ENCODES

Urinary loss of antithrombin III (and other regulators) plus increased hepatic procoagulants causes a hypercoagulable state: DVT, PE, and especially renal vein thrombosis (most classic with membranous). Consider prophylactic anticoagulation when albumin is very low.

BOARD TRAP

Renal vein thrombosis presenting as flank pain + worsening proteinuria in a membranous patient — anticoagulate, don't just up immunosuppression.

13. Infection — Encapsulated Organisms

WHAT YOU SEE

Warning, ↓ antithrombin/IgG

WHAT IT ENCODES

Urinary loss of immunoglobulins (IgG) and complement raises infection risk, classically with encapsulated organisms like Streptococcus pneumoniae (spontaneous bacterial peritonitis in children). Vaccinate against pneumococcus.

BOARD TRAP

Nephrotic children with abdominal pain — think spontaneous pneumococcal peritonitis, not just appendicitis.

14. Accelerated Atherosclerosis

WHAT YOU SEE

Clogged-artery icon

WHAT IT ENCODES

Chronic hyperlipidemia and hypercoagulability accelerate cardiovascular disease in persistent nephrotic syndrome. Statins are indicated for sustained dyslipidemia.

BOARD TRAP

CV risk in nephrotic syndrome is driven by the dyslipidemia/hypercoagulable milieu — a complication, treated, not the disease itself.

15. Vitamin D Binding Loss / Bone Disease

WHAT YOU SEE

Bone icon, center-left

WHAT IT ENCODES

Loss of vitamin D-binding protein in urine lowers 25-OH vitamin D, causing hypocalcemia and metabolic bone disease. Also lose transferrin (microcytic anemia) and thyroxine-binding globulin.

BOARD TRAP

The hypocalcemia is from urinary vitamin-D-binding-protein loss, not parathyroid disease — total calcium also drops with albumin.

16. AKI — Intravascular Volume Depletion

WHAT YOU SEE

Banner, bland sediment/no RBC casts

WHAT IT ENCODES

AKI can occur from intravascular volume depletion (aggressive diuresis), interstitial edema, or renal vein thrombosis; sediment is bland (no RBC casts), distinguishing it from nephritic AKI.

BOARD TRAP

Bland urine sediment with AKI in a nephrotic patient points to prerenal/volume/RVT causes — RBC casts would mean a nephritic process instead.

17. Treatment — RAAS + Statin + Diuretic

WHAT YOU SEE

Treatment pharmacy shelf

WHAT IT ENCODES

Backbone: ACE inhibitor/ARB to lower proteinuria, loop diuretics + salt restriction for edema, statin for dyslipidemia, anticoagulation when high thrombotic risk, plus cause-specific immunosuppression (e.g., steroids for MCD/primary FSGS, rituximab for PLA2R membranous).

BOARD TRAP

Don't immunosuppress before identifying the cause — diabetic and amyloid nephropathy worsen, not improve, with steroids.
